

Project Communication Strategy

Communication Plan: Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Team Standup	Daily	Slack / Teams	Dev Team	Task Sync & Blockers
2	Progress Update	Weekly	Email / Dashboard	Team & Mentor	Status Review
3	Issue / Bug Discussion	As Needed	GitHub Issues / Call	Developers	Bug Resolution
4	Stakeholder Review	Bi-Weekly	Google Meet	Mentors / Team	Milestone Feedback
5	Final Demo Rehearsal	Once	Live Presentation	All Hands	Final Validation

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	API Rate Limits & Latency in Gemini Model	Implemented response caching and optimized prompt structure for faster processing.
2	Cross-platform budget breakdown inconsistencies	Standardized output JSON formats and strictly defined schema parsing logic.
3	Integration delays between Frontend & Backend	Established structured API endpoints using FastAPI/Flask and maintained clear API specs.

PocketSmart AI: Your Smart Budget & Recommendation Assistant

Project Description

Managing budgets across different life needs—like home decor, event planning, or jewelry shopping—can be overwhelming due to the wide variety of products, platforms, and price ranges. PocketSmart AI addresses this challenge through a GenAI-powered, cross-platform recommendation system that delivers personalized, budget-based suggestions for products and services. Using Gemini 1.5 Flash Pro, PocketSmart AI analyzes user preferences, budgets, and contextual needs to generate curated recommendations from popular platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, and more. The platform helps users plan interior designs, parties, and even select matching jewelry for special occasions—all within their defined budgets. PocketSmart AI

transforms budgeting into an intelligent, user-friendly experience through its smart planners, adaptive forms, and recommendation engine that bridges multiple categories and e-commerce ecosystems.

Key Scenarios

Scenario 1: Home Interior Planning with Smart Budget Allocation

Users begin by selecting the **Home Budget Planner**. After entering their budget, they specify room types (e.g., Living Room, Kitchen, Bedroom) and quantities for lights, ceiling fans, dining tables, etc. Gemini 1.5 Flash Pro processes this data and recommends cost-effective options for each category across platforms like IKEA and Amazon. Recommendations are balanced across functionality, style, and price, ensuring users can decorate their spaces beautifully without overspending.

Scenario 2: AI-Based Party Budget Planning

In the **Party Planner**, users input their total budget, guest count, event type, and venue details. Based on these inputs, the AI allocates the budget proportionally across catering, decoration, and entertainment, sourcing options from vendors listed on Swiggy, Zomato, and accommodation services like OYO. The system tailors suggestions based on event type (e.g., birthday, corporate, wedding), helping users organize a successful event without financial guesswork.

Scenario 3: Jewelry Recommendations for Occasions

The **Jewelry Budget Planner** allows users to enter a budget, select the occasion, and define style preferences. Users can optionally upload an outfit image for the AI to consider color coordination and aesthetics. Gemini 1.5 Flash Pro analyzes the data and offers jewelry options from platforms like Amazon and Flipkart, tailored to occasion and style. This ensures that users receive elegant, matching jewelry suggestions that fit both their personal style and financial plan.

Architecture Overview

PocketSmart AI is an intelligent, modular platform designed to generate personalized, budget-aware recommendations across domains like home interiors, party planning, and jewelry selection. It integrates a lightweight backend (Flask/FastAPI) with Google's Gemini 1.5 Flash Pro, a powerful multimodal AI model that processes user inputs (text, numbers, and images) to deliver smart recommendations.

Core Technologies

- **Flask / FastAPI:** Backend service layer for request handling, routing, session control, and API validation.
- **Gemini 1.5 Flash Pro:** Generative AI model used for understanding budget context, analyzing outfit images, generating structured suggestions, and recommending platform-specific products.
- **Third-party Platform Integration:** Dynamic generation of shopping links for Amazon, Flipkart, IKEA, Zomato, Swiggy, Bluestone, and Tanishq.
- **Frontend (HTML/CSS + JS):** Interactive user interface for input collection, live validation, and structured output rendering.

Component-Wise Architecture

Component	Description
Frontend UI	Web interface built with HTML/CSS/JS that allows users to input budgets, preferences, and images. Displays tailored AI-driven recommendations in structured cards.
Backend Service	Manages user sessions, authentication, and routing (/generate-home, /generate-party, /generate-jewelry), and handles communication with the AI engine.

Component	Description
Gemini AI Layer	Processes multimodal inputs using Gemini 1.5 Flash Pro to extract budget parameters, perform reasoning, and format recommendations into JSON schemas.

Pre-requisites

- **1. IBM Cloud / Google Cloud Account Setup:** Access to Cloud Console and Gemini API keys.
- **2. AI Model Access & Configuration:** Google AI Studio API access configured.
- **3. Python Basics:** Python 3.9+ environment setup.
- **4. FastAPI / Flask Framework:** Core routing and server libraries installed.

Project Execution Workflow

MILESTONE 1: Gemini AI Initialization

In this foundational stage, the core AI infrastructure required to power PocketSmart AI is established. Authenticated access to Gemini APIs is configured to perform text reasoning and image analysis.

- **Activity 1.1:** Set up Google Cloud / Google AI Studio account and enable Gemini API access.
- **Activity 1.2:** Configure API keys, environment variables (GOOGLE_API_KEY), and security permissions.
- **Activity 1.3:** Validate API connectivity using sample Python test scripts for both text-only and image+text multimodal prompts.

MILESTONE 2: Core Functionalities Development

Build the core backend logic, prompt templates, and recommendation helpers for each planner module.

- **Activity 2.1:** Initialize FastAPI/Flask dependencies, set up CORS middleware, password hashing, and static file directories.
- **Activity 2.2:** Implement core routing functions:
 - /generate-home: Generates interior recommendations (furniture, decor, lighting) structured by rooms and item counts.
 - /generate-party: Suggests catering, venue, decoration, and entertainment tailored to guest counts and event type.
 - /generate-jewelry: Accepts text and optional outfit image uploads to return style-matched accessories.
- **Activity 2.3:** Implement authentication endpoints (/login, /register, /logout).
- **Activity 2.4:** Implement session management and JWT authentication tokens (/token, /session-info).

MILESTONE 3: Backend - FastAPI/Flask Integration

Connect core utilities with web application endpoints and history logs.

- **Activity 3.1:** Define structured POST endpoints (/home-budget, /party-budget, /jewelry-budget).
- **Activity 3.2:** Create modular architecture and setup endpoint for fetching recommendation details (/recommendation-details/{id}).
- **Activity 3.3:** Implement recommendation history endpoints (/history, /recommendation-history).
- **Activity 3.4:** Setup background server cleanup tasks and startup handlers using Uvicorn.

MILESTONE 4: UI Development

Develop responsive web templates using HTML5, CSS3, and Jinja2 styling for a clean user dashboard.

- **Activity 4.1:** Build interactive planner forms, dashboard widgets, and card-based recommendation displays.

MILESTONE 5: Testing & Optimization

Perform end-to-end testing, prompt optimization, and edge-case handling.

- **Activity 5.1:** Validate real-world budget scenarios across all three planning modules. Ensure budget limits and currency calculations are strictly maintained.

Conclusion

Each module—from registration to recommendations—is designed to simplify decision-making and enhance the shopping or planning experience through automation and personalization. Whether it's finding the right wall art within budget or organizing a last-minute party, PocketSmart AI acts as a smart digital assistant, saving time and effort while optimizing outcomes. With its seamless fusion of generative AI and web technologies, PocketSmart AI stands as a practical example of how AI can enhance everyday decision-making.